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GEOPHYSICS, VOL. XXII, NO. 2 (APRIL, 1957), PP. 384–397, 9 FIGS.

POLAR MOVEMENT AS INDICATED BY ROCK MAGNETISM*

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ABSTRACT

Measurements of remanent magnetism have been carried out by geophysicists both in this country and in the British Isles in a broad sampling of Paleozoic sediments. The direction of magnetization of these rocks indicates that the positions of the earth's magnetic poles were quite different in Paleozoic times from their positions today. Since it is felt that the magnetic poles are closely tied in with the spin axis of the earth, the locations of the magnetic poles, as averaged over space and time, should indicate the locations of the geographic poles. Determinations of magnetization of Mississippian concretions and other sediments of chemical origin in the Llano uplift area of Texas were made in our laboratory, and indicate pole positions in Carboniferous times consistent with observations made by Belshe in England and by Runcorn in Arizona. Cambrian pole locations, as found from samples in the same area, are included and compared with positions from other work. A few interesting data are shown from the Sawatch formation of Cambrian Age in Colorado.

INTRODUCTION

Polar movement has been discussed in geologic literature for many years in an attempt to explain evidence of climatic changes on a world-wide basis in the distant past. Wegener has even gone so far as to postulate relative movement or drift between the continents. One of the more modern tools in the study of paleoclimates is the oxygen-isotope-ratio method developed by Urey (1948) and his colleagues. There is, however, a limitation here in that only sea water temperatures are considered. It is now beginning to appear that the most direct approach to the determination of the positions of the poles in the past is offered by measurements of the direction of remanent magnetism of rocks of various ages. Already a quantity of data has been accumulated from both igneous and sedimentary formations.

In the case of igneous rocks, it has been demonstrated by various observers that the rock becomes magnetized in the direction of the earth's field as it cools from the molten state through the Curie point. Hospers (1955), on the basis of his own work and the work of others, has concluded that the magnetic pole has remained at or very near the present geographic pole for at least back into the Miocene. It is assumed in all of this work that if averages of the direction of magnetization of samples covering sufficient geologic time (say thousands of years) are taken, the deviations due to secular variation of the magnetic field are reduced so that the position of the earth's magnetic pole agrees with that of the rotational or geographic pole. Thus, Hospers concludes that the geographic poles have not moved appreciably since the end of the Eocene. The close relationship between the magnetic and the geographic poles is not surprising on the basis

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of the modern dynamo theories of earth magnetism of Elsasser (1946 and 1947) and Bullard (1949).

Another interesting observation made on the measurements of igneous rocks is that about half of the Cenozoic lava flows (*Handbuch der Physik*, 1956) show a magnetization in a direction reverse to the usual direction. These data have brought about an argument as to whether or not the magnetic poles of the earth can reverse. There is evidence, as pointed out by Nagata (1952), that some igneous rocks can actually be magnetized in a direction opposite to that of the field applied as they cool from the paramagnetic state. Also, there is other evidence that the field of the earth itself has reversed. Although a mass of data has already been accumulated, this question needs and will undoubtedly receive further study.

In the case of sediments, which in general can be more accurately dated than igneous formations, Runcorn (1955a) and his group at Cambridge as well as others have found very striking evidence that the poles have moved, and in particular that the Paleozoic pole locations are quite different from their present locations. Graham (1949) at the Department of Terrestrial Magnetism in Washington set forth basic criteria for determining the stability of remanent magnetism with respect to geologic time. These criteria were concerned with sedimentary rocks, but the stability question must be applied to both igneous and sedimentary rocks since it is vital to know whether a rock was magnetized at the beginning of its history or later. Graham found that the directions of magnetization in a Silurian formation, folded near the end of the Paleozoic, were quite scattered, but that the directions were brought into good agreement by flattening the fold with the corresponding rotation of the magnetic vectors in space. Graham concluded that the magnetization had been stable at least since the folding at the end of the Paleozoic. The other important finding was that the direction of magnetization was quite different from the direction of the earth's present magnetic field. These data will be referred to later.

Although the manner in which igneous rocks are magnetized is fairly well accepted, the mechanism is not always clear in the case of sedimentary rocks. At times it has been suggested that detrital particles align themselves in the earth's field as they settle in the water, or as they rest in the ooze on the bottom. However, this brings up the question as to whether the particles realign themselves upon striking the bottom, with a resulting change in dip direction of the magnetization. We have recently published some results (Martinez and Howell, 1955b) obtained with chemical sediments for which it would be expected that the magnetization took place during the chemical deposition. This should avoid a possible difficulty encountered with detrital particles.

OBSERVATIONS IN THE LLANO UPLIFT AREA

We have sampled the Barnett formation of Mississippian age in the Llano uplift area of central Texas, the locations being represented by squares in Figure

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1. We feel that we have obtained a satisfactory sampling in both time and space as the extreme sample locations are roughly 70 miles apart. The samples were obtained largely from ellipsoidal calcareous concretions, as well as from limestone and shale beds. Figure 2 is a picture of a typical Barnett concretion.

Figure 3 shows a Schmidt projection of the directions of magnetization obtained from individual plugs from the nine locations. The plots show the direction of the north-seeking vector on a lower hemisphere; i.e., all vectors are directed down. The azimuth from north is shown in the conventional manner while the angle made with the vertical is shown as a function of the radius. Thus, a number of the vectors are nearly horizontal in a southeasterly direction. It is very interesting that location *J* reveals a magnetization reversed from that of most of the data. Also several points from locations *D*, *G*, and *H* fall near the direction of the earth's present field, and thus indicate recent magnetization. Some of the other points indicate varying amounts of remagnetization, especially from data taken from some of the shale at location *F*. It is for this reason that the two representative vectors are selected as shown by the two large circles. It is felt that the large cluster of points at the extreme southeast represents a more stable direction of magnetization than vectors closer to the earth's present magnetic field.

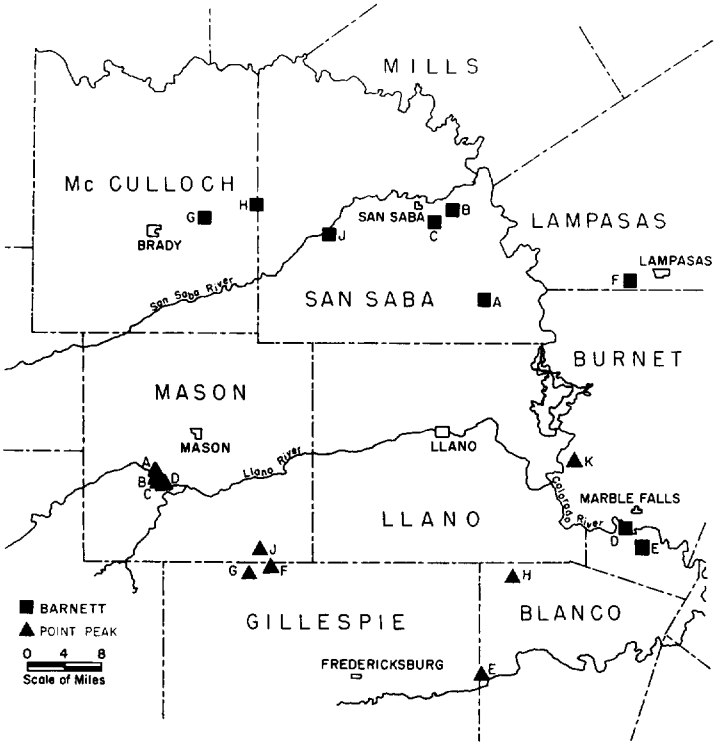


FIG. 1. Map showing sample locations in the Llano uplift area of Central Texas.

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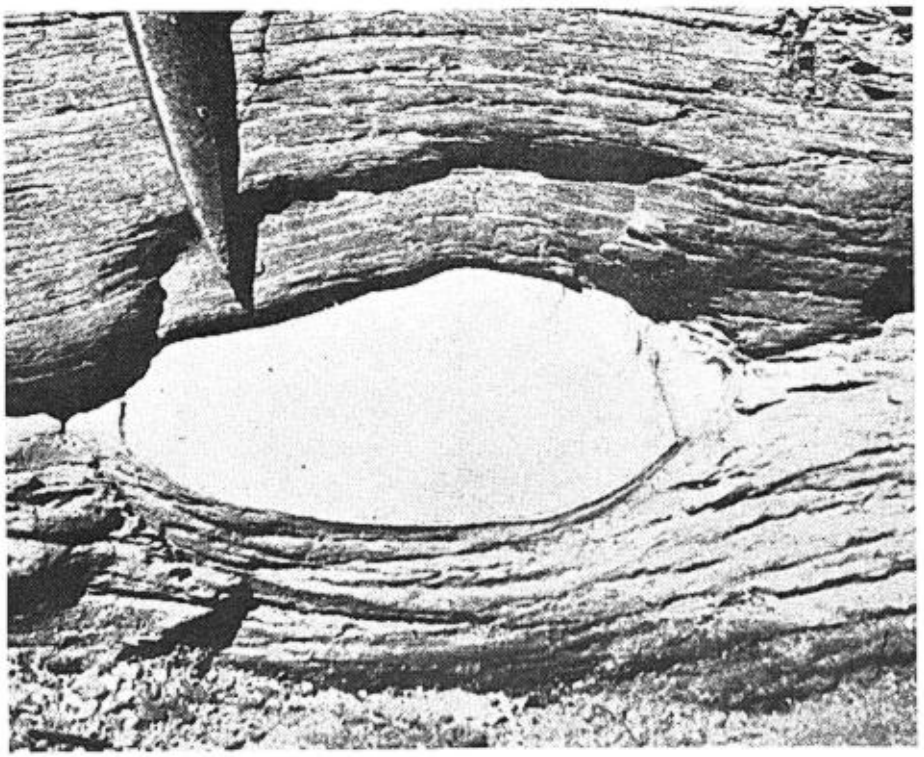


FIG. 2. Typical concretion in Barnett formation of Mississippian age. The point of a geologist's hammer is shown for scale.

LOCATIONS OF THE MAGNETIC POLES

Figure 4 shows the world-wide location of the poles computed from these directions. The computations are made on the usual assumption of a centered-dipole geomagnetic field. The "starred" circles represent the positions of the poles computed from the vector to the southeast, while the plain circles show the positions from the reverse vector. The positions are given below.

	South Pole	North Pole
"Starred" Poles	Lat. 41° N Long. 128° E	Lat. 41° S Long. 52° W
Location *J* Poles	Lat. 42° S Long. 38° W	Lat. 42° N Long. 142° E

The "starred" locations are fairly close to those published as computed from a vector-sum average, from which some of the points were excluded. The "J" poles are the same as previously. As to the map, it is of interest that our pole locations agree quite well with those of Belshe (Runcorn, 1955b) as indicated, his samples coming from Carboniferous rocks in England. The sample location in each case is indicated by the symbol enclosing an "L." Agreement is shown with Runcorn's (1955b) poles from data of Pennsylvanian rocks obtained in the Arizona area. We

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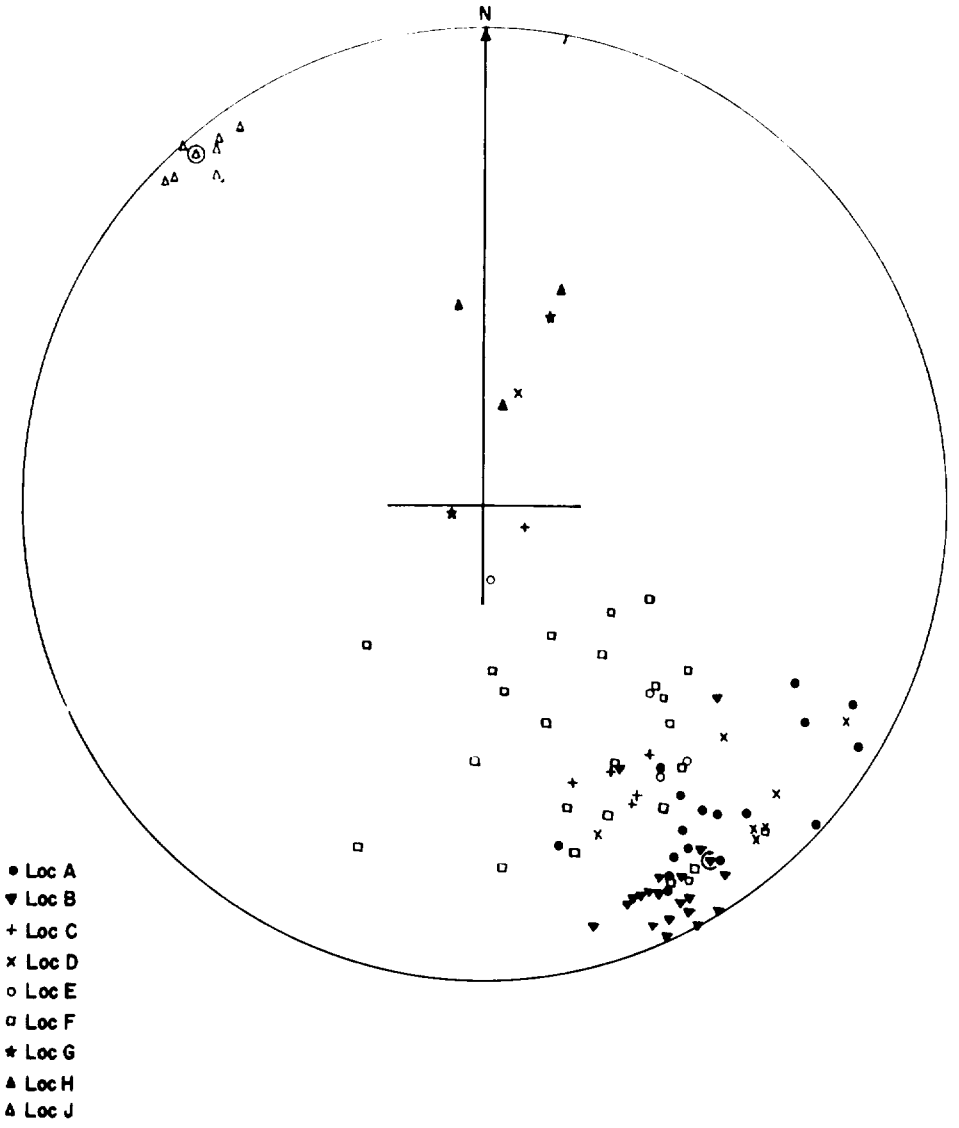


FIG. 3. Barnett magnetic vectors plotted on lower hemisphere of Schmidt net.

feel that this agreement is sufficient evidence that very little continental drift between England and our continent has occurred. It might be pointed out that recently Runcorn (1956) postulated a drift of about 20 degrees mainly on the basis of Paleozoic data. Another point of interest on the map is the equator corresponding to our "starred" poles. This equator passes very close to the areas of Carboniferous and Permian reefs in Texas and New Mexico; thus, the reefs

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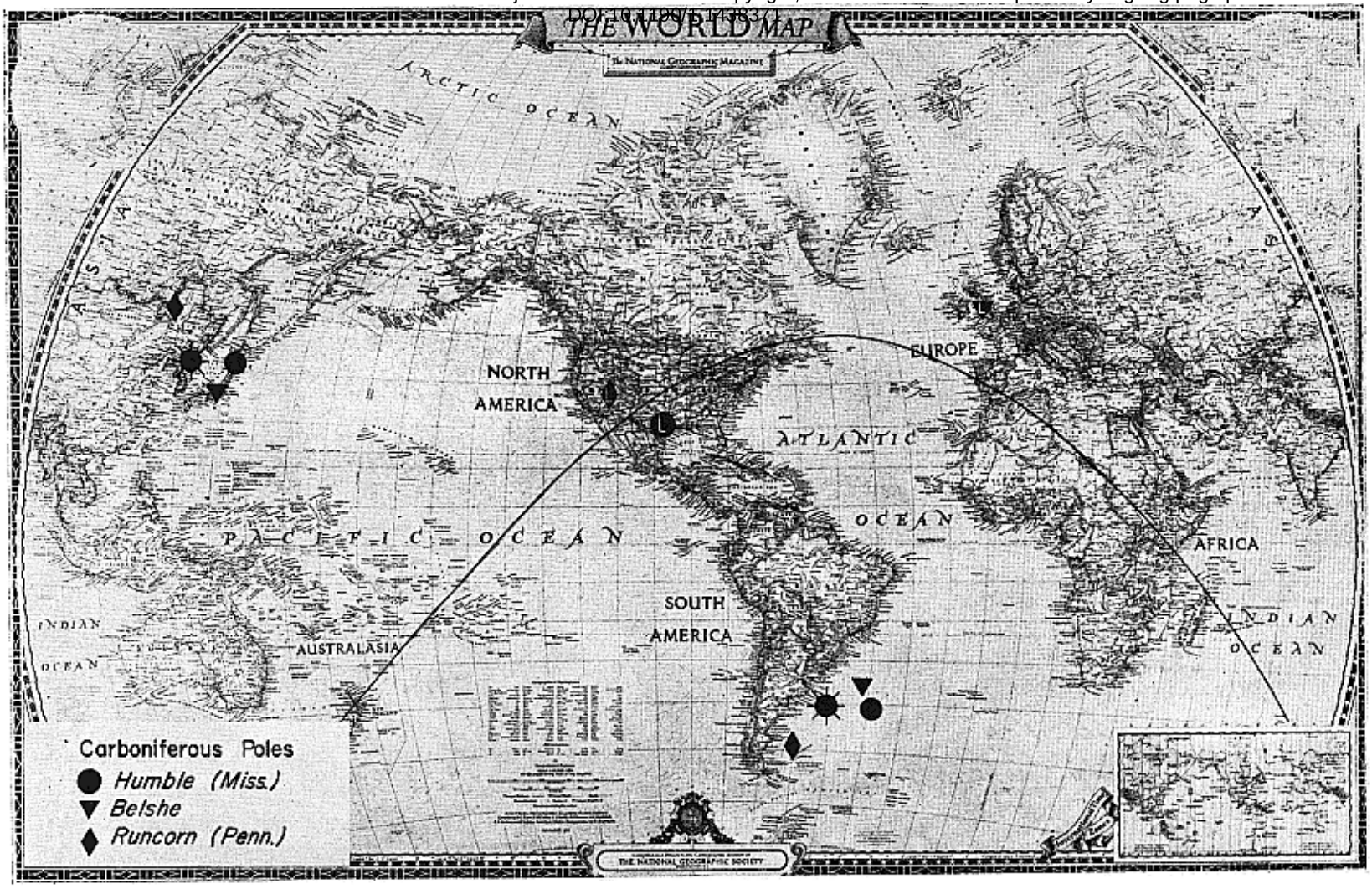


FIG. 4. Map of Carboniferous poles and equator. Sample locations are indicated by symbols enclosing "*L*."

probably enjoyed a favorable environment of sunlight and warm water at the time they were formed.

FURTHER OBSERVATIONS IN THE LLANO UPLIFT AREA

We have also made measurements on upper Cambrian samples obtained from the Point Peak member of the Wilberns formation in the Llano uplift area. These locations are represented by triangles in Figure 1. The samples in general are siltstone. Figure 5 shows a plot of the magnetic vector directions on a Schmidt net. The few points falling on the upper hemisphere are dotted in. Also, it may

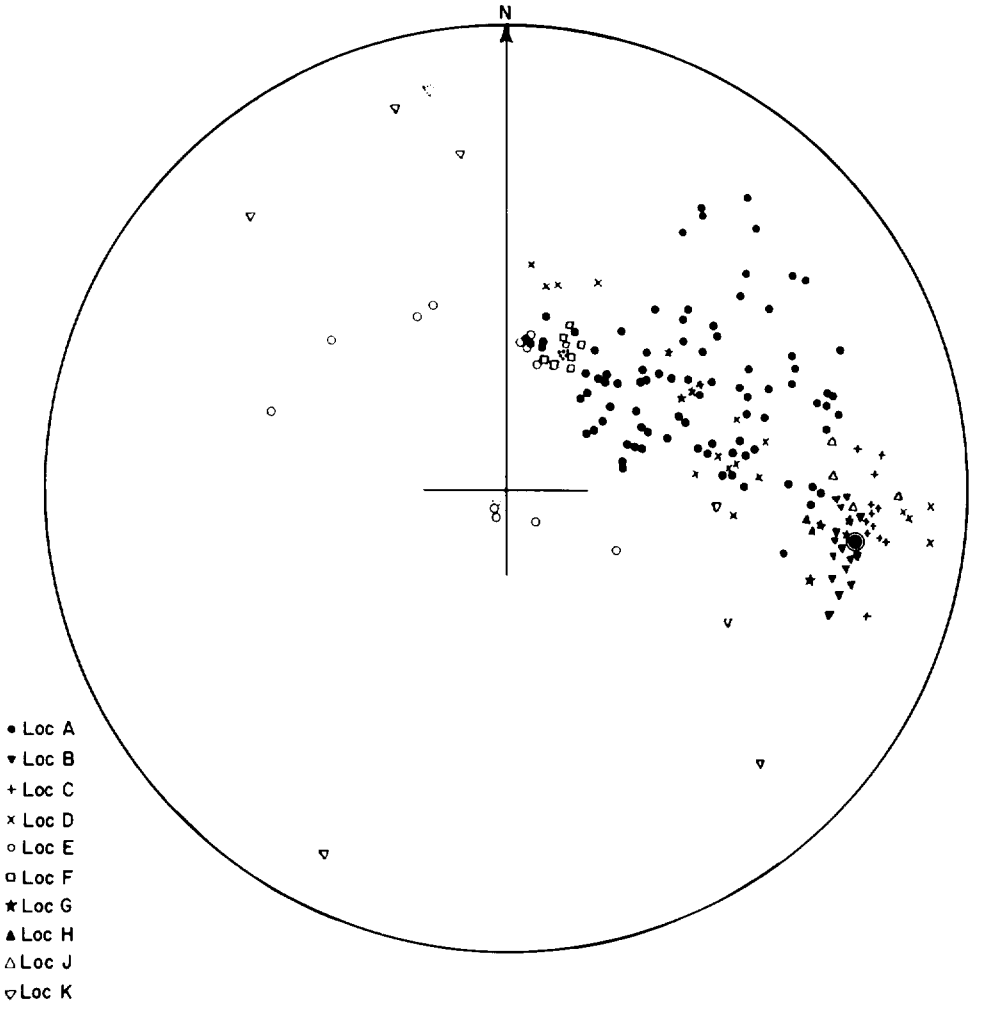


FIG. 5. Point Peak (Cambrian) magnetic vectors, plotted on Schmidt net, all points being on lower hemisphere with exception of dotted points on upper hemisphere.

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be mentioned that in general the coordinate system is tied in with the plane of the bed; i.e., stratum dips have been removed. Steep dips occurred at only one location, *K*, and here they ranged from 30 degrees to 60 degrees. The vectors are erratic for this location. On the whole, the Point Peak samples do not exhibit the same magnetic stability as the Barnett. However, a representative vector has been chosen near the head of the points and marked with a multiple circle.

LOCATIONS OF THE MAGNETIC POLES

The pole locations computed from this vector are shown in Figure 6. The positions are given below.

South Pole	North Pole
Lat. 0° Long. 158° E	Lat. 0° Long. 22° W

These pole locations, designated by gray triangles, are in fairly good agreement with those made by Creer (Day and Runcorn, 1955) from samples in Great Britain. The locations by Day and Runcorn (1955) from Arizona samples are in lesser agreement. We have, however, seen no data published for this case. The pole positions obtained by Du Bois (1955) from the upper Keweenawan in Michigan are included, although the Keweenawan is generally classified as pre-Cambrian. There has, however, been some question concerning the age of the upper Keweenawan in Michigan. The magnetic data indicate the possibility that it is of Cambrian age, or at least very near Cambrian, as pointed out by Du Bois.

We have also included our pole locations (squares) from the Sawatch formation in Colorado. The vector data are shown on the Schmidt plot in Figure 7. Although we do not have widespread data from this formation, having sampled only two locations, the data suggest the interesting possibility that this Cambrian rock may have been magnetized near the end of Paleozoic time. The rocks were determined, on the basis of thin section examination, to be principally sandy dolomites containing abundant glauconite. Much hematite is present and occurs in irregular masses, interstitial filling between dolomite crystals, a fine dust incorporated in the lattice of the dolomite crystals, and as a partial replacement of glauconite and in some cases quartz. The glauconite, except as noted above, is unaltered. The hematite seems to have formed concurrently with dolomitization.

One possible mechanism for late postdepositional magnetization might be formation and magnetization of the hematite in a late dolomitization. Another possible mechanism might be remagnetization of the rock in the presence of stresses related to formation of the Ancestral Rockies.

Figure 8 shows the pole data obtained from Triassic-Permian formations by various observers. In the bottom part of the map there are shown pole positions computed by Graham (1955) from data for individual sample locations made by Clegg, Almond and Stubbs (1954) at London University. The locations are in Triassic outcrops in England. Also shown are poles for the individual locations made by Graham (1955) on Triassic-Permian outcrops in Maryland and in the

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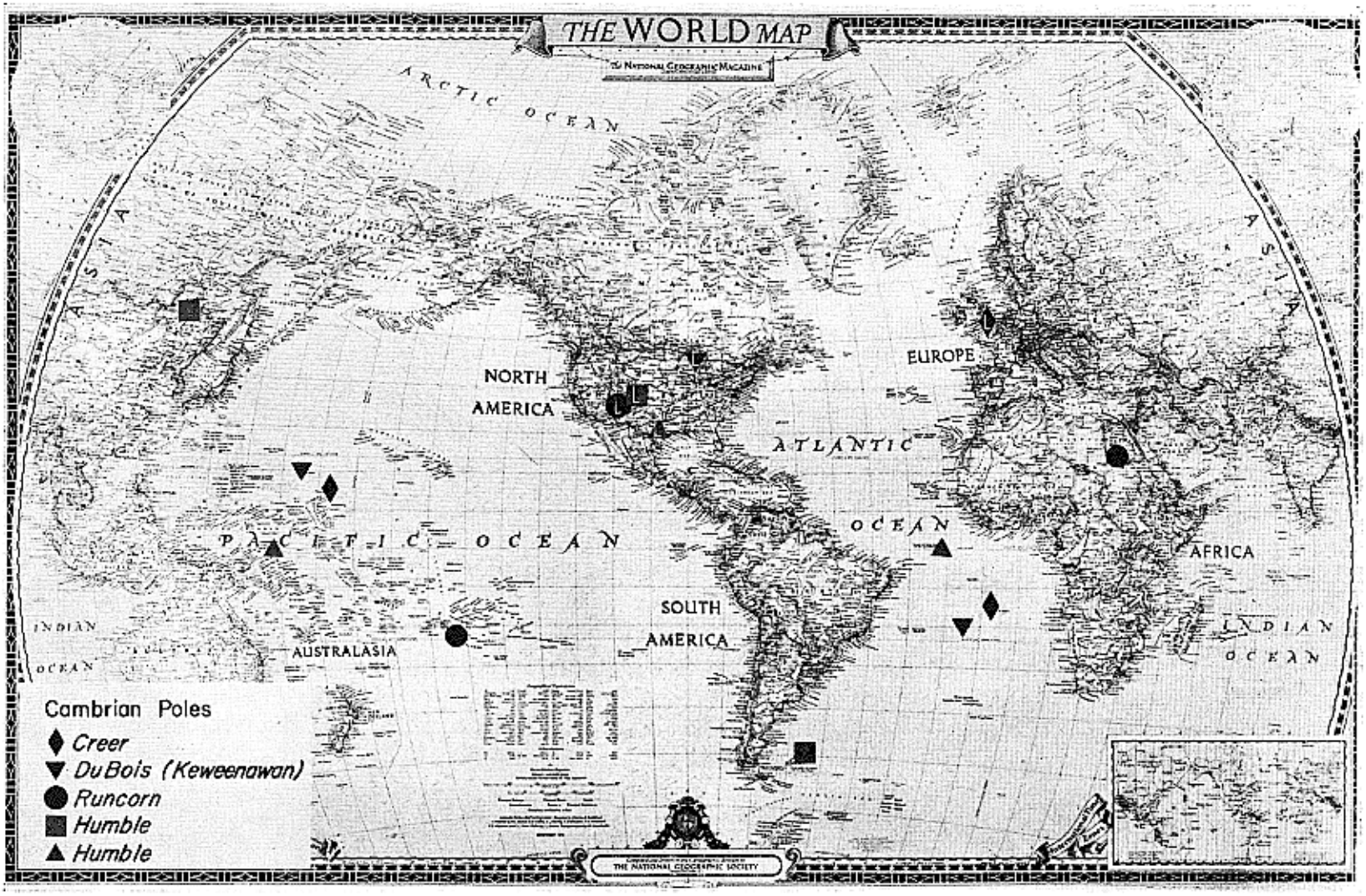


FIG. 6. Map of Cambrian pole locations.

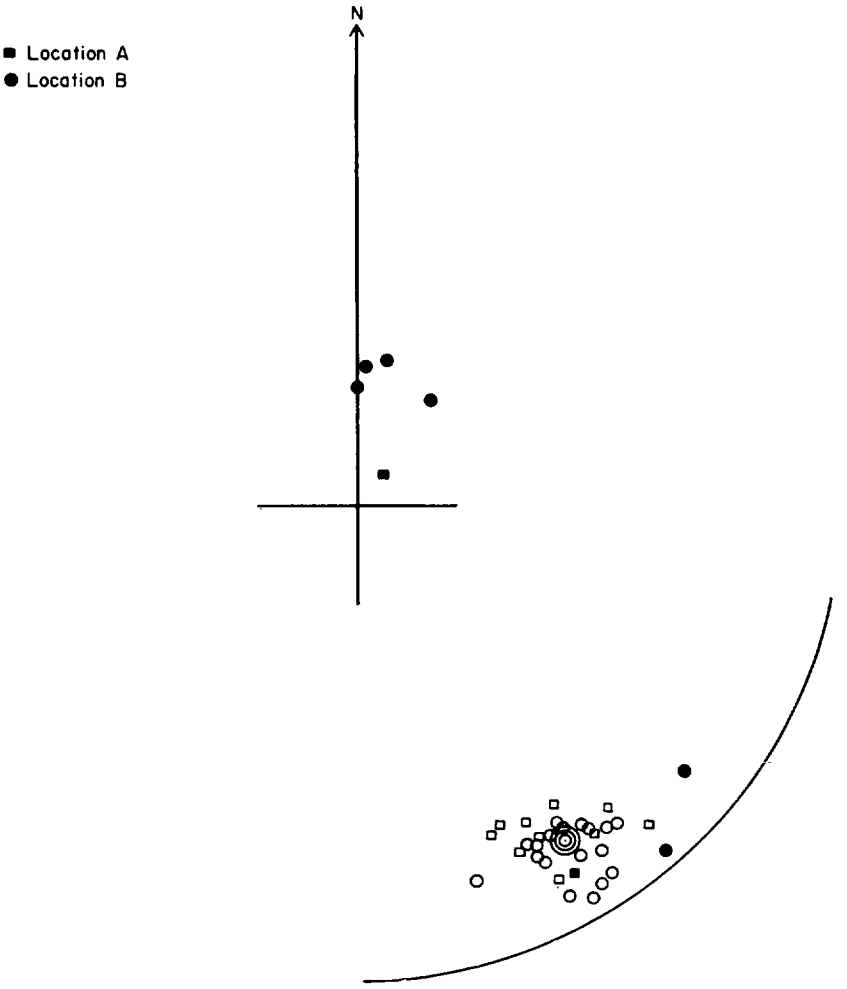


FIG. 7. Sawatch magnetic vectors plotted on Schmidt net. Solid symbols represent lower hemisphere points; open symbols, upper hemisphere.

Arizona-New Mexico area. These poles are in some instances North and in others, South. From the estimated mean positions of the poles, as shown in the upper left hand corner of Figure 8, the agreement in the positions of the poles is very good between data from England and from the United States. The Triassic pole located by Creer (Day and Runcorn, 1955) represents another determination using the same measurements of Clegg as given above. The Triassic and Permian poles located by Runcorn (1955b) from measurements on rocks obtained in the Utah-Arizona area are in fair agreement. Doell (1955) also obtained data from the same Permian formation in Arizona as sampled by Runcorn. The pole loca-

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FIG. 8. Map of Triassic-Permian pole locations.

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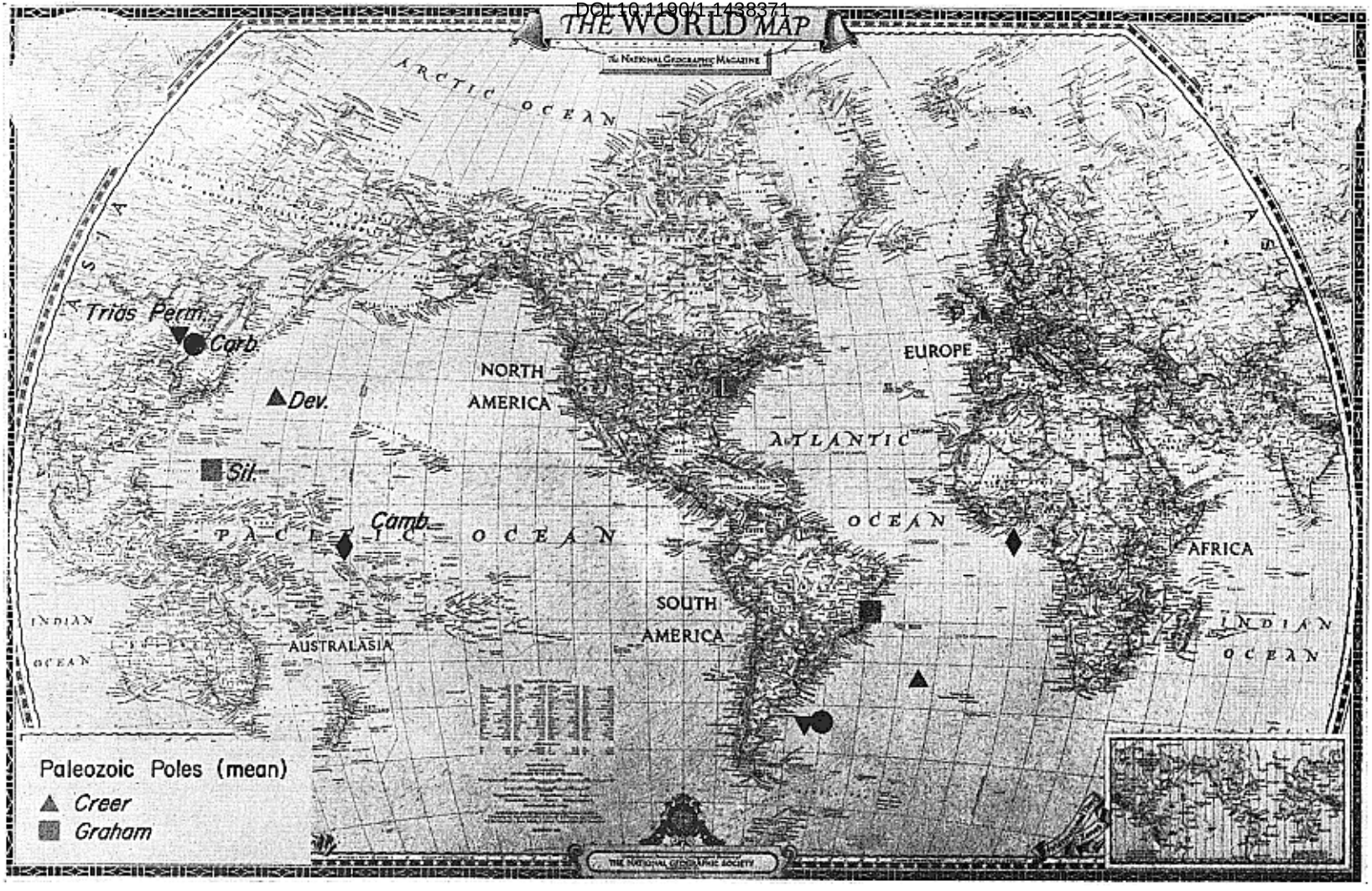


FIG. 9. Map of mean pole locations for the Paleozoic.

tion in the northern hemisphere as determined by Doell is somewhat to the north of that determined by Runcorn.

In Figure 9 approximate averages are shown for the positions of the poles in Paleozoic times (the Triassic being lumped in with the Permian). In obtaining a mean for the Cambrian poles, only the determinations by Runcorn, by Creer, and by us, using the Point Peak data, are considered. Two other ages, not shown before, are included. These are the pole locations from Graham's 1949 data for the folded Silurian at Pinto, Maryland, and the Devonian pole locations made by Creer (Runcorn, 1955b) from British samples. It should be remembered that all of these paleozoic data were obtained from sedimentary formations.

On the whole, there seems to be a suggestion that the poles moved in a progressive fashion during Paleozoic time; certainly their positions are quite different from those of today. Over-all, there is fairly good agreement between the pole locations obtained from Great Britain and from our country with the possibility of only a small displacement as suggested by Runcorn.

CONCLUSIONS

In conclusion, the geophysicists now have a powerful tool for studying changes which not only affect paleoclimates but also may produce crustal shear patterns on a world-wide basis as postulated by Vening-Meinesz (1947). We who are engaged in geophysical prospecting are beginning to realize more and more the importance of the remanent magnetism of igneous rocks in magnetic surveys. In the past, induced magnetism has received the major portion of the emphasis, but we know that in some cases the remanent magnetism of an igneous rock may exceed the amount of induced magnetization by one-hundred fold. Also, with more and more paleomagnetic data the diagnostic properties of remanent magnetism are becoming apparent. As an example, Lundbak (1956) in Denmark has pointed out the possibilities of analyzing paleomagnetic data from a combination of gravity and magnetic surveys. Certainly it could be very useful to know the direction of remanent magnetism and the inferred age of the igneous rocks in a basin. Finally, we must emphasize that we are in the early stages of this very interesting field of paleomagnetism.

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